

Preliminary Research Report

Rare-Earth Free Permanent Magnets and 2D Ising-Model Simulation

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1. Exploration of Key Terms

After reading the review paper, I identified several key terms that are important for understanding the properties that define a good permanent magnet. These terms include:

Saturation Magnetization (M_s) - The maximum magnetization a material can reach when an external magnetic field aligns all the magnetic moments in the material. A high M_s is desirable for a strong permanent magnet but high M_s alone is not sufficient to create a permanent magnet.

Coercivity (H_c) - The magnitude of the reverse magnetic field required to bring back a magnetized material back to 0 net magnetization. High coercivity is a key characteristic desired in rare earth free magnets as the material can hold its magnetic field as it has high resistance to demagnetization.

Remanence (M_r) - The magnetization remaining after the external magnetic field is removed. A useful permanent magnet should have high M_r .

Magnetocrystalline Anisotropy - The tendency of a material's magnetization to prefer certain directions over others. Strong anisotropy makes it more difficult for the magnetization to rotate away from its preferred direction, which can increase coercivity.

Curie Temperature (T_c) - The temperature above which thermal fluctuations destroy the ferromagnetic properties of the material. Above T_c , the material becomes paramagnetic and can no longer maintain its magnetization. A T_c below room temperature can make a permanent magnet not useful for certain applications even if it has other properties that make it a good permanent magnet.

Crystal structure - The repeating arrangement of atoms within a crystal. Structures such as body centered cubic (bcc), face centered cubic (fcc), and tetragonal structures can produce very different magnetic behavior even when using the same materials.

Exchange interaction - The interaction that causes neighboring magnetic moments to influence each other's alignment. In a ferromagnet, it generally favors parallel alignment.

2. Preliminary Materials Survey

Fe-Co

Fe-Co alloys have very high M_s but its anisotropy is very low. The current research attempts to distort the normally cubic lattice into a tetragonal lattice which theoretically greatly increases its anisotropy while maintaining its high M_s . However, the tetragonal configuration is difficult to maintain in bulk as strained Fe-Co films become thicker, the lattice tends to relax back toward its more stable cubic structure.

Fe-Ni (Tetrataenite)

Fe-Ni suffers from the same disadvantages that Fe-Co does. It has high M_s but low anisotropy. A particularly good phase of this material is L10-Fe-Ni, also known as Tetrataenite, which has good M_s and anisotropy. It also has the benefit of having high T_c . The difficulty comes from forming the ordered L10 phase within the material. Researchers are now thinking of solutions to try to synthesize this in bulk quickly and economically while having its desirable properties.

Fe₁₆N₂

A particular phase of Fe-N magnets being explored is the α'' -Fe₁₆N₂ as it has high theoretical M_s and anisotropy and its main difficulty is how to process the material. The material is metastable and the amount of nitrogen has to be carefully controlled during synthesis. Too much or too little nitrogen will make it form other undesirable phases. Another major weakness is that its metastability makes bulk processing difficult, since conventional fabrication methods often require high temperatures that can destabilize and destroy the desired phase.

Fe₂-B

Fe₂-B magnets are highly desirable because of its chemical stability, very high T_c and mechanical hardness but it behaves more like a soft magnet rather than a hard magnet due to its low coercivity caused by its unfavourable anisotropy. Researchers are therefore investigating ways to increase the anisotropy in various ways such as substituting Fe for Co.

Fe₂-P

Fe₂-P has very high anisotropy due to its hexagonal lattice structure but its T_c is very low and impractical for many applications. Researchers are therefore trying to modify the lattice to increase its T_c such as adding Co, Si and B into the lattice without affecting its desired magnetic traits. The difficulty comes from trying to balance all these substitutions to find a good composition as different compositions can create unwanted phases or reduce its anisotropy.

The common challenge among these rare-earth free materials is that no option currently combines all of the properties of an ideal permanent magnet. Fe-Co provides excellent M_s but lacks sufficient anisotropy. L₁₀-Fe-Ni has promising magnetic properties but is difficult to chemically order. α'' -Fe₁₆N₂ has very high potential magnetization but poor phase stability. Fe₂-P based materials require careful compositional modification to remain magnetic above room temperature, while Fe₂-B has excellent thermal stability but insufficient coercivity. The goal of current research is therefore not simply to maximize magnetization, but to also optimize their magnetization, coercivity, phase stability, and microstructure.

3. Computational Model

These materials show that achieving a strong permanent magnet requires more than simply having high M_s . A major challenge is understanding what allows a magnetized state to remain stable when an opposing magnetic field is applied. To explore this behavior in a simplified system, I began using a 2-dimensional Ising model to study the competition between interactions among neighboring magnetic moments and an external magnetic field. The model is not intended to represent a specific material directly, but instead provides a simple framework for studying magnetization reversal, hysteresis, and coercivity.

The current model uses a square 2D Ising lattice in which each spin has two allowed states, $\text{spin} = +1$ or -1 . The energy contains a nearest neighbor exchange term and an external field term:

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - H \sum_i s_i$$

- J is the nearest-neighbor exchange coupling; $J = 1$ is used as the energy scale.
- H is the externally applied magnetic field.
- Normalized magnetization is $M = (1/N) \sum_i s_i$, ranging from -1 to $+1$.
- Periodic boundaries giving every site four nearest neighbors.
- The model has been tested on 4×4 , 10×10 , 20×20 , and finally 50×50 lattices for the final analysis.

4. Monte-Carlo Implementation

A Metropolis Monte Carlo update has also been implemented. Each sweep attempts $L \times L$ randomly selected spin updates. Energy lowering flips are accepted; for $T > 0$, energy increasing flips are accepted with probability $\exp(-\Delta E/T)$

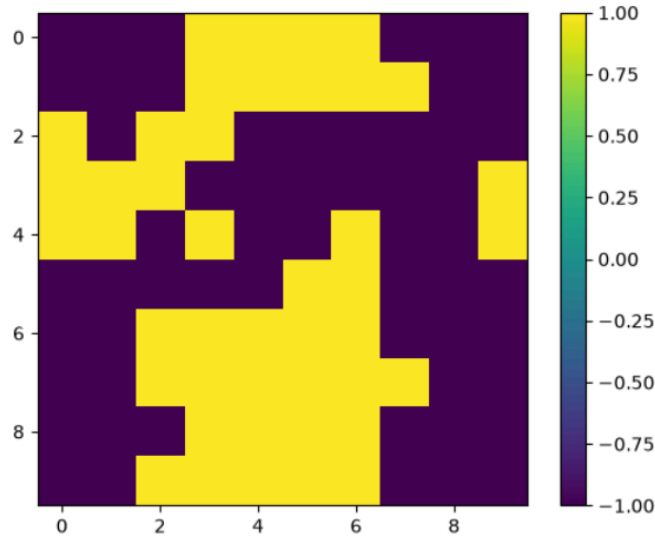


Figure1: the spin state of each site for a 10x10 Lattice after 10 metropolis sweeps with initial conditions $J=1$, $T=3$, $H=0$ and all sites initialized to $s_i = +1$

5. 0-Temperature Baseline and Hysteresis

At $T = 0$, the simulation accepts spin flips only when they do not increase the energy. Sweeping the applied field from positive to negative and back to positive produces a nearly square hysteresis loop.

For a spin with four aligned neighbors, reversal changes the exchange contribution by $8J$ while an opposing field contributes $-2|H|$. The local energy change is therefore $\Delta E = 8J - 2|H|$. With $J = 1$, the instability condition $\Delta E = 0$ occurs at $|H| = 4$.

The $T = 0$ simulation switches at approximately $H/J = \pm 4$, which provides an analytical sanity check for the implementation.

Temperature: 0

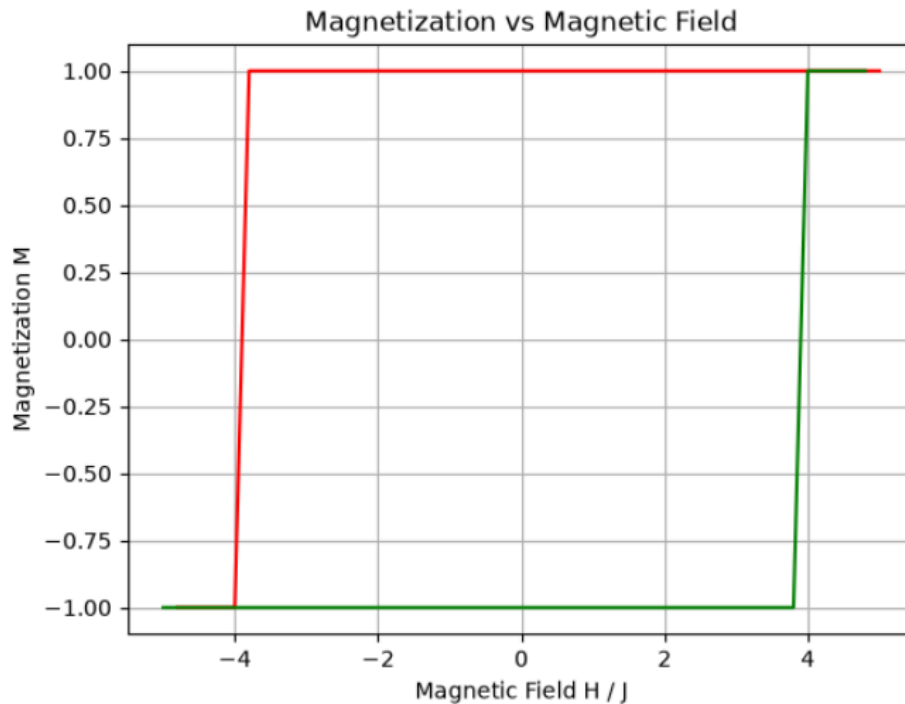
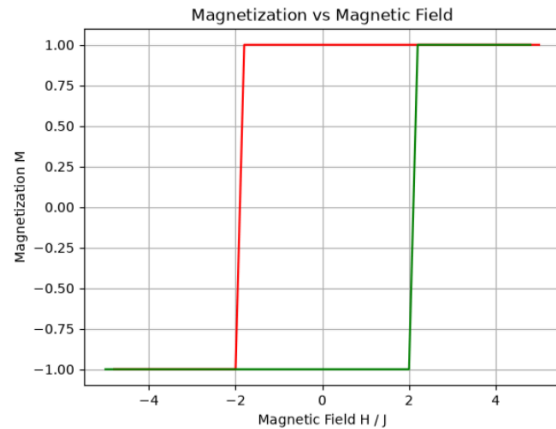


Figure2: Hysteresis Loop for $T=0$ where initial $H=5$ and goes to $H=-5$ and back to $H=5$

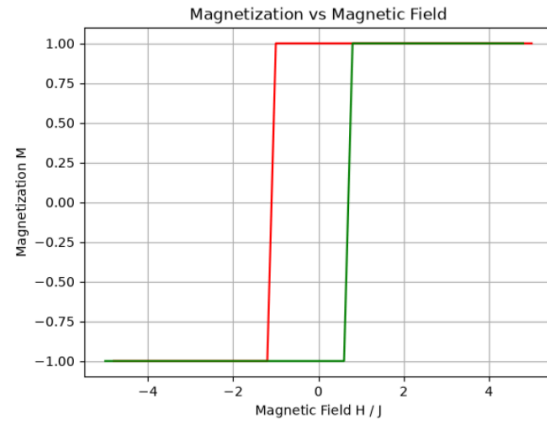
6. Temperature Dependence of Hysteresis

I repeated the magnetic field sweep at several nonzero temperatures. The figures below show hysteresis loops for $T = 0.5, 1.0, 1.5, 2.0$, and 2.5 . As temperature increases, thermal fluctuations make it easier for spins to flip even when doing so temporarily increases the energy. As a result, the magnetization begins to reverse at smaller opposing magnetic fields and the width of the hysteresis loop decreases.

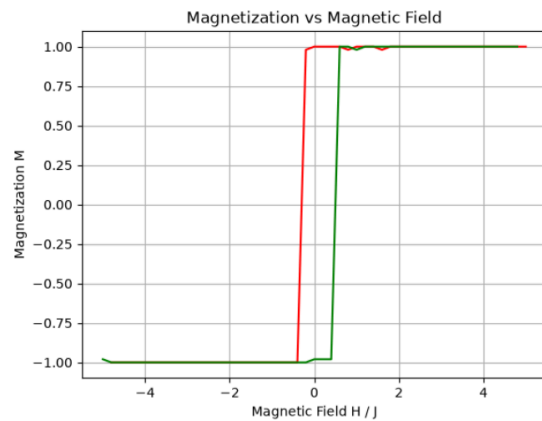
Temperature: 0.5



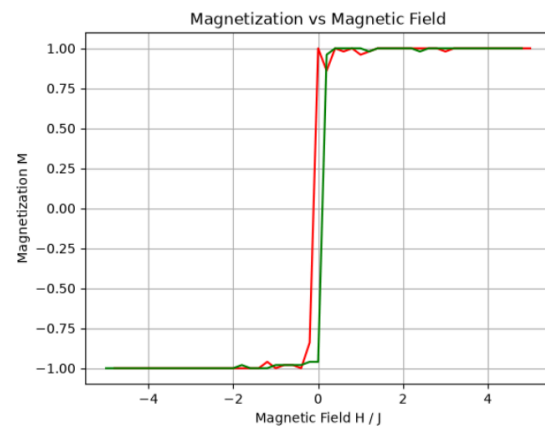
Temperature: 1.0



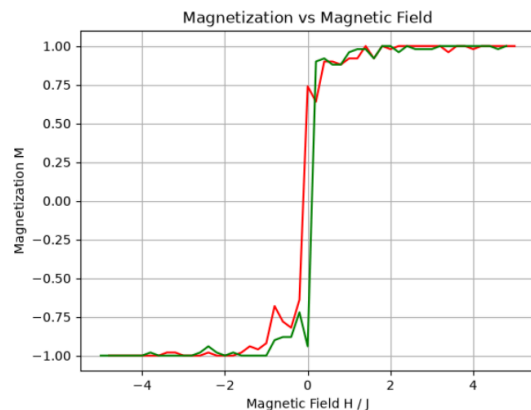
Temperature: 1.5



Temperature: 2.0

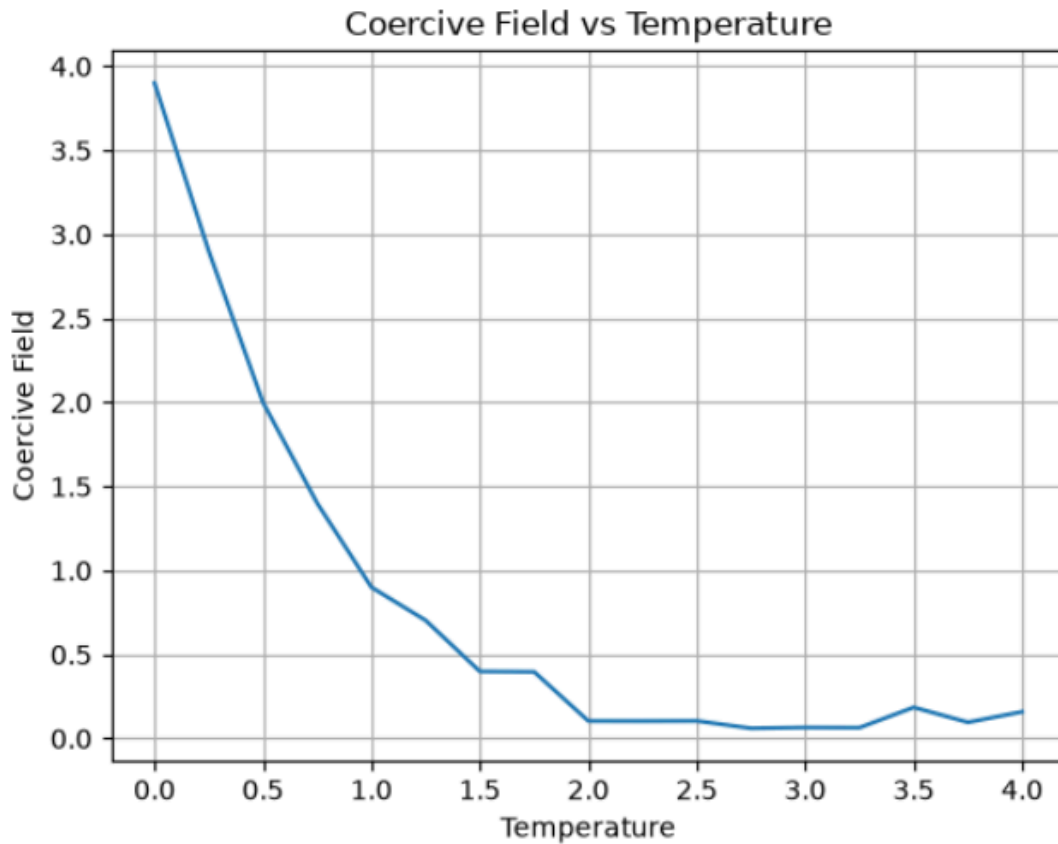


Temperature: 2.5



At lower temperatures, the loops remain relatively square and the system stays close to full magnetization until reversal occurs. As the temperature approaches $T = 2.0$ - 2.5 , the transition becomes less abrupt and larger fluctuations in magnetization appear. This indicates that thermal energy is increasingly competing with the exchange interaction that keeps neighboring spins aligned.

These results suggest that the coercive field decreases as temperature increases. However, these results are preliminary and can depend on lattice size, number of Monte Carlo sweeps, field step size, and the rate at which the magnetic field is changed.



To quantify this trend, I extracted the coercive field from each hysteresis loop by finding the magnetic field where the magnetization crosses zero using Interpolation. The resulting coercive field as a function of temperature is shown above. The overall decrease in H_c with increasing temperature is consistent with the narrowing of the hysteresis loops observed above.

7. Limitations of the Model and Future Work

The current 2D Ising model is a highly simplified representation of magnetic behavior and does not directly represent a real permanent magnet material. In the model, every lattice site has only two possible spin states and every neighboring pair uses the same exchange coupling J . In real materials, magnetic moments can point in many directions, and the strength of the exchange interaction can vary depending on the atoms involved, their spacing, crystal phase, defects, and interfaces.

The model also uses periodic boundary conditions whereas real magnets are finite and contain surfaces, grains, grain boundaries, defects, and multiple magnetic domains. Magnetization reversal in real permanent magnets can therefore occur through processes such as domain nucleation and domain wall motion, which are not fully represented by the current model.

Despite these limitations, the model provides a useful starting point for understanding how exchange interactions, external magnetic fields, and temperature compete to produce hysteresis and coercivity. Future work could gradually introduce more realistic effects such as spatially varying exchange coupling, defects, different boundary conditions, or interactions that better represent magnetic anisotropy and domain behavior.